

- 7 At time ts , a particle P is projected with speed 40 ms^{-1} at an angle θ above the horizontal from a point O on a horizontal plane and moves freely under gravity. The greatest height achieved by P during its flight is $H \text{ m}$ and the corresponding time is $T \text{ s}$.

(a) Obtain expressions for H and T in terms of θ . [2]

[illegible]

During the time between $t = T$ and $t = 3$, P descends a distance $\frac{1}{4}H$.

(b) Find the value of θ . [4]

This image shows a full page of white paper designed for handwriting practice. It features 18 evenly spaced, horizontal dashed lines that run across the entire width of the page. The lines are thin and black, providing a guide for letter height and placement. There are no margins, text, or other markings on the page.

- (c) Find the speed of P when $t = 3$.

[3]

[illegible]